

International dynamic marketing capabilities and international performance during the pandemic: a study of export manufacturers from Poland

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Abstract

Purpose – This paper contributes to studies on the relationship between dynamic capabilities (DCs) and performance by showing how domain-specific DCs – international dynamic marketing capabilities (IDMCs) – affect the international performance of exporting firms in the context of extreme environmental dynamism – during the COVID-19 pandemic.

Design/methodology/approach – The authors focus on a sample of 277 exporting manufacturers from the post-transition economy of Poland. The authors use hierarchical multiple regression analysis to test this study's hypotheses.

Findings – This study's findings show that deployment of IDMCs by export manufacturers in the context of environmental jolts contributes to better performance, and this relationship is mediated by adaptation to foreign markets and product development capability. Additionally, this study's results reveal that the significant and positive indirect effect of IDMCs on international performance (through mediators) is, however, weakened under conditions of extreme environmental dynamism.

Research limitations/implications – The limitations pertain to the cross-sectional nature of this study and the research sample, characterised by the dominance of export manufacturers of final products, the dominance of manufacturers operating in the business-to-business sector, or in the business-to-business and business-to-customer sectors simultaneously.

Practical implications – The study provides suggestions to managers on how to build resilience in international markets during turbulent times. These activities involve investments in IDMCs that support activities centred around product development and adaptation to foreign markets.

Originality/value – The novel construct of IDMCs is introduced and operationalized. The study empirically tests the direct and indirect relationship between IDMCs and performance contingent upon extreme environmental dynamism. The results demonstrate the boundary conditions for the effectiveness of these domain-specific DCs in such a research setting.

Keywords International dynamic marketing capabilities, International performance, Environmental dynamism, Transforming countries, Poland

Paper type Research paper



1. Introduction

The dynamic capabilities perspective (DCV) has been increasingly adopted by international business scholars to explain how firms overcome the challenges of foreign expansion and improve their international competitiveness (Weerawardena *et al.*, 2015; Wójcik and

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Ciszewska-Mlinaric, 2021; Zahra *et al.*, 2022). DCV is suitable for analysing the effects of international firm-level behaviour, since both dynamic capabilities (DCs) and the internationalization process involve adaptation to contingencies and continuous reconfiguration and development of resources (Verbeke, 2022). DCs, defined as “the firm’s ability to integrate, build, and reconfigure internal and external competences to address rapidly changing environments” (Teece *et al.*, 1997, p. 516), are considered especially useful in sensing, seizing and responding to opportunities and threats, as well as achieving abnormal performance (Teece and Leih, 2016). It has been argued that the recent coronavirus pandemic crisis reflected a discontinuous environmental change. On the one hand, it increased uncertainty to a greater extent than any other crisis before, threatening the existence of many businesses and posing new challenges to international expansion in some industries (Wenzel *et al.*, 2020). On the other hand, it also created opportunities in other industries through the digitalization of business models and the acceleration of internationalization (Amankwah-Amoah *et al.*, 2021).

Scholars have noted that a firm’s competitiveness depends strongly on the development of domain-specific DCs (Andreeva and Ritala, 2016). In contrast to generic DCs, which are universally applicable and related to organization-wide learning, domain-specific DCs pertain to specific functional domains (Hine *et al.*, 2014). In particular, the application and development of dynamic marketing capabilities (DMCs) during foreign expansion has recently received increased interest from international marketing scholars (Buccieri *et al.*, 2020; Mitrega *et al.*, 2021; Xu *et al.*, 2018). DMCs are customer-focused cross-functional capabilities that enable the effective accessing of market knowledge about customers’ needs and contribute to delivering customer value (Mitrega, 2019). In an international context, DMCs are a firm’s ability to build, integrate and reconfigure strategic marketing acumen to effectively identify and deliver value to international markets (Weerawardena *et al.*, 2015). Numerous studies analysing the role of DMCs in the international process indicate that possession of DMCs, accompanied by entrepreneurial orientation and innovativeness, is necessary for the creation of value for international customers and improving international performance (e.g. Buccieri *et al.*, 2020; Falasca *et al.*, 2017; Konwar *et al.*, 2017; Weerawardena *et al.*, 2015). However, despite the progress made by this research, the empirical results on the DMC–performance relationship are still inconclusive. This ambiguity is partly associated with the inconsistent operationalization of DMCs and the research context across studies. For instance, Barrales-Molina *et al.* (2014) indicate that extant studies on DMCs focus on chosen marketing aspects related to either new product development, customer relationship management or brand management. Falasca *et al.* (2017) focus on supply chain management, while Konwar *et al.* (2017) concentrate on marketing intensity. In particular, Barrales-Molina *et al.* (2014) raise that the literature conceive multiple understandings of underlying processes (e.g. sensing, seizing, coordinating, integrating, reconfiguring), specific components (related to learning and knowledge management), marketing enabler processes (related to customers, interorganizational relationships, brand, distribution) of DMCs. Consequently, these diversities in the literature raise difficulty in understanding the processes and mechanisms through which DMCs operate and affect performance. Furthermore, prior empirical studies were conducted on relatively homogenous samples of high-tech international new ventures (INVs) and small and medium-sized enterprises (SMEs) operating predominantly within developed or large developing economies, such as India and China (Kowalik *et al.*, 2020). Moreover, although the literature points to the different effects of institutional conditions (Lundan and Li, 2019) and levels of environmental dynamism on the DC–performance relationship (Karna *et al.*, 2016; Fainshmidt *et al.*, 2016), prior empirical studies were conducted in relatively stable international environments. These studies found a direct impact of DMCs on performance (Konwar *et al.*, 2017) and product innovation performance (Falasca *et al.*, 2017), but also that the development of DMCs affecting firm performance is

stronger under environmental turbulence (Buccieri *et al.*, 2020). However, we lack full understanding of how DMCs affect the consequences under discontinuous environmental change. Taken together, a lack of understanding of the environmental conditions shaping the consequences of DMCs' deployment in international markets, not only in terms of performance, but also in terms of resource reconfiguration modes, calls for further research (Buccieri *et al.*, 2020; Mitrega *et al.*, 2021; Weerawardena *et al.*, 2015).

Accordingly, in this study we address these gaps and ask the question: *How do specific DCs, namely international dynamic marketing capabilities (IDMCs), affect firms' international performance during the COVID-19 pandemic?* In doing this, we extend the existing concept of dynamic marketing capabilities (DMCs) to the international context, labelling them as "international dynamic marketing capabilities (IDMCs)", and investigating their direct and indirect effects on international performance. We postulate that this relationship is mediated by adaptation to foreign markets and product development capability, and moderated by environmental dynamism. We therefore adopt a configurational approach to the DC–performance relationship by including both organizational and external factors (Fainshmidt *et al.*, 2019; Wilden *et al.*, 2016).

The study seeks to contribute to the existing literature in two ways. First, it seeks to advance our understanding of the role of IDMCs in international performance. In particular, it adds to prior conceptualizations of dynamic marketing capabilities in the international context by testing the conceptual model of the relationship between the novel concept of IDMCs and international performance, mediated by adaptation to foreign markets and product development capability, and moderated by extreme environmental turbulence. The *novelty of the concept* relates to two aspects: (1) the application of a new name for DMCs in the international context – i.e. IDMCs; and (2) operationalization of the IDMC construct, which builds on prior research on DMCs (Mitrega, 2019) but is adjusted to the specificity of internationalization. Second, it aims to contribute to international marketing research through the empirical context of our study. We argue that it is unique and thus provides valuable theoretical insights by (1) considering extreme environmental turbulence (the coronavirus pandemic), often overlooked in studies on DC–performance relationship, and (2) focussing on resource-constrained companies from the post-transformational economy of Poland. In other words, our research sample offers an opportunity to test existing theories relating to international marketing and dynamic marketing capabilities in a new context – that of a Central and Eastern European (CEE) economy (Mitrega *et al.*, 2021; Peng *et al.*, 2018). Hoskisson *et al.* (2013) and Kostova and Hult (2016) argue that the CEE region is peculiar and differs not only from developed economies, but also from other "emerging" economies due to different initial institutional conditions and rapid growth. Furthermore, the international position of export manufacturers from Poland in global value chains is characterized by vertical specialization which, combined with the relatively high import intensity of exports, as well as demand and supply disturbances caused by the pandemic, creates risks of decreased sales and markups (Hagemejer and Mućk, 2019; Hagemejer *et al.*, 2022).

2. Theoretical background

Dynamic capabilities (DCs) advance our understanding of the one of fundamental problems in strategy and international business research: firms' capacity to respond to changing environments. The need to study DC development has arisen in recent decades due to the requirements posed by environmental dynamism, and has become even more important in the face of the current environmental jolt (the coronavirus pandemic). DCs are defined as "the capacity of an organization to purposefully create, extend, or modify its resource base" (Helfat *et al.*, 2007, p. 4) to match a changing business environment (Teece, 2007). Furthermore, Teece (2007) proposes that DCs operate through three fundamental mechanisms

(microfoundations): sensing, seizing and transforming. Sensing allows the external environment to be scanned so as to identify opportunities. Seizing relies on processes and routines related to accessing external knowledge, and its subsequent absorption and integration with the firm's existing processes, routines and resources. Seizing refers to governance structures and mechanisms, including delegation, decentralization, coordination, reward system, decision-making protocols and processes (making investments in organizational assets and deciding on organizational structure). Transforming constitutes organizational processes and routines through which resources and operating capabilities may be recombined.

DCs are viewed as higher-order capabilities that enable the organization to reconfigure resources and develop ordinary lower-order capabilities, such as new product development (Eisenhardt and Martin, 2000; Pavlou and El Sawy, 2011). A widely accepted view is that the impact of DCs on firm performance is an indirect, two-step process through reconfiguration of ordinary lower-order capabilities which, in turn, have a direct effect on a firm's performance (e.g. Hine *et al.*, 2014; Kalubanga and Gudergan, 2022; Wilden *et al.*, 2016). Hine *et al.* (2014) distinguish two types of DCs: "dynamic learning capabilities" and "dynamic functional capabilities". Similarly, Andreeva and Ritala (2016) recognize dynamic "generic" and dynamic "functional" capabilities. While the dynamic "generic"/"learning" capability has universal application (it is not restricted to a certain function), the dynamic "functional" capability, also referred to as "domain-specific" DCs, is specialized in a specific organizational function (e.g. marketing, procurement, sales). Therefore, the effect of domain-specific DCs on performance takes place through alteration of functional resources and ordinary lower-order capabilities. In this vein, the literature specifies four various modes: leveraging, creating, accessing and releasing (Danneels, 2011; Eisenhardt and Martin, 2000). In this context, the systematic assessment by Pezeshkan *et al.* (2016) of 89 empirical studies revealed that marketing DCs were found to contribute significantly to improved performance in the majority of empirical tests exploring this relationship (i.e. 80% of the analysed articles), although this was true to a lesser extent when combined with the interaction with high environmental dynamism. This overall positive relationship was mediated in the analysed studies by resources and lower-order capabilities.

2.1 Environmental dynamism as a moderator of the dynamic capabilities–performance relationship

There is an abundance of studies on the relationship between DCs and performance. However, the empirical findings suggest that the impact of DCs on outcomes is complex and indirect (Fainshmidt *et al.*, 2016; Wang *et al.*, 2015). For instance, developing and deploying DCs is associated with necessary resource commitments. Therefore, DCs may also lead to the deterioration of market position or firm performance in the short-run (Winter, 2003) because an irrelevant set of resources is developed (Wang and Ahmed, 2007). Furthermore, the DC–performance relationship involves multiple mediators and moderators (Bitencourt *et al.*, 2020; Schilke *et al.*, 2018), with environmental contingencies, such as institutional conditions (Lundan and Li, 2019; Fainshmidt *et al.*, 2016) or environmental dynamism, being the most studied (Pezeshkan *et al.*, 2016). In general, although studies on the DC–performance relationship have made some progress, some unresolved issues still exist (Schilke *et al.*, 2018). One of these is associated with the applicability and effectiveness of DCs in moderately vs rapidly changing environments (Peteraf *et al.*, 2013). Therefore, a major question in the DCs research stream refers to the environmental conditions under which DCs generate most positive results for the firms that develop them ("the boundaries to the effectiveness of DCs"; Verona and Zollo, 2011, p. 536).

The original assumption is that DCs are especially effective in rapidly changing environments (Teece *et al.*, 1997), and several empirical studies have confirmed this (e.g. Barrales-Molina *et al.*,

2014; Drnevič and Kriauciunas, 2011). In this vein, Li *et al.*'s (2019) study confirmed the effectiveness of DCs in high-velocity environments. A more universal view postulates a positive DC–performance relationship across the entire spectrum of environmental dynamism, i.e. low and high (e.g. Protogerou *et al.*, 2012). Yet, another view suggests that in volatile environments, DCs are less useful or not at all in improving performance and level of competitiveness (Eisenhardt and Martin, 2000). Finally, a view that has more recently gained in importance aligns with the contingency hypothesis (Girod and Whittington, 2017). Studies in this area suggest a non-linear DC–performance relationship, postulating a weaker effect of DCs on a firm's competitiveness in low levels of environmental dynamism because of insufficient opportunities to cover the cost of their development (Drnevič and Kriauciunas, 2011; Helfat *et al.*, 2007; Winter, 2003). For instance, Schilke (2014) found that this relationship has an inversed U-shaped curve, such that DCs have “the strongest positive impact on competitive advantage under intermediate environmental dynamism, being thus weaker in stable and highly dynamic contexts” (p. 197). In their meta-analysis conducted on a set of 82 samples (reported in 79 studies), Fainshmidt *et al.* (2016) found that DCs contribute more to performance in developing economies than in developed economies, suggesting that DCs are more valuable in unstable than stable environments. As Girod and Whittington (2017, p. 1131) note, “the positive returns to reconfigurations in dynamic environments suggest that the advantages of continuous small changes in those conditions are greater than those of the larger”.

To summarize, while the relationship between domain-specific DCs, such as customer or supplier relationships (Pavlou and El Sawy, 2011; Wang *et al.*, 2015), and performance has been extensively studied in moderate and high velocity environments, we have limited understanding and knowledge of the effect of dynamic marketing capabilities on performance during a discontinuous change in the environment, i.e. an environmental jolt, that is a sudden and unprecedented event.

2.2 The role of dynamic capabilities in international business

While research on the role of DCs in a firm's internationalization has gained momentum in recent years, the field is still in the nascent stage of development (Verbeke, 2022), and remains limited and fragmented (Zahra *et al.*, 2022). First, the abundance of conceptual works contrast with the relative scarcity of empirical studies. A direct effect of this state is that DCs in international business are de-contextualized (Lundan and Li, 2019), i.e. they have not been systematically measured neither in different institutional nor environmental contexts, creating gaps in the possible types of DCs, their constituent elements and dimensions (Zahra *et al.*, 2022). Nevertheless, these studies suggest that DCs, including domain-specific ones which contribute to international expansion, may have their own peculiarities making them different in their nature and constituent elements from domestic ones (Morgan *et al.*, 2018; Verbeke, 2022; Zahra *et al.*, 2022). Therefore, our understanding of domain-specific DCs remains underexplored, especially those related to marketing activities, their relationship with other capabilities, and their combined effect on international performance (e.g. Bucciari *et al.*, 2020; Morgan *et al.*, 2018; Weerawardena *et al.*, 2015). Second, empirical studies have focused predominantly on the process of development of DCs for international purposes, rather than on the effects of DC application. Here, special focus has been placed on learning processes and their effects on enhancing DCs (e.g. Riviere *et al.*, 2018). Third, the majority of studies on the role of DCs in internationalization are centred around problems with governance of business units, i.e. coordination of the internationally dispersed activities of multinational corporations (e.g. Teece, 2014; Zahra *et al.*, 2022), or high-tech international new ventures and born globals (e.g. Weerawardena *et al.*, 2015). Therefore, traditional exporters have been overlooked (Wójcik and Ciszewska-Mlinarič, 2021). Fourth, the empirical studies have been limited to developed or large developing economies, such as India and China

(Kowalik *et al.*, 2020). Fifth, the empirical studies have been conducted during relatively stable economic periods, and environmental dynamism has so far been reflected in the conditions in emerging economies. Therefore, the period of the COVID-19 pandemic constitutes a natural context for empirically testing the DCs–performance relationship during internationalization under extremely dynamic environmental conditions.

2.3 *The role of international dynamic marketing capabilities in international expansion*

The literature on strategy recognizes dynamic marketing capabilities as domain-specific DCs that play an important role in international markets, due to their contribution to improved firm performance (e.g. Tan and Carlos, 2015). According to Fang and Zou (2009), the distinguishing feature of DMC versus generic DCs is the former's orientation towards customer value. In turn, Mitrega (2019) emphasized that market knowledge implemented to reconfigure company resources distinguishes DMCs from other DCs the most. In this sense, DMCs, in contrast to generic DCs, represent domain-specific DCs pertaining to specific functional domains (Hine *et al.*, 2014). DMCs are defined as a company's mechanisms through which marketing resources are reconfigured in response to market changes (Falasca *et al.*, 2017). These mechanisms are cross-functional processes (integrating various departments) and rely on market knowledge (Barrales-Molina *et al.*, 2014; Morgan, 2012). Furthermore, the literature suggests that in order to have a positive effect on a company's performance, DMCs have to be integrated with the company's innovation activities and new product development (Mitrega, 2019). There are significant theoretical premises indicating that application of such domain-specific DCs in international markets may potentially enhance international performance. The literature suggests that sales are positively affected by DMCs, but that in the context of foreign expansion, DMCs are expected to be enhanced by the possession of other organizational capabilities (Mitrega, 2019). For instance, a study conducted by Mitrega *et al.* (2021) on a sample of 590 Polish manufacturing companies tested and found a direct effect of DMCs on product and market expansion.

The construct of DMCs is relatively new and is still in the process of being scientifically specified (Mitrega *et al.*, 2021). It can be argued that most works examine DMCs in the context of firms' activities on international markets and with regards to transforming or emerging economies (e.g. Bucciari *et al.*, 2020; Falasca *et al.*, 2017; Fang and Zou, 2009; Konwar *et al.*, 2017; Mitrega, 2019; Mitrega *et al.*, 2021). This appears to be related to the greater uncertainty and ambiguity associated with companies' operations in foreign markets as compared to domestic markets (e.g. Konwar *et al.*, 2017; Weerawardena *et al.*, 2015). Therefore, we propose that DMCs studied in an international context should be named IDMCs. It should be noted that the international context of some other research (e.g. Mitrega, 2019; Morgan *et al.*, 2018) into DMCs means that the term IDMCs could just as well have been used in these studies. Indeed, this is how it was treated when Kim and Lim (2022) used the term IDMCs to refer to these studies specifically.

The dynamic nature of IDMCs is associated with the fact that they are concerned with marketing resource reconfiguration and capability enhancement for responding to international market environment efficiently and promptly (Fang and Zou, 2009). In other words, IDMCs refer to the ability to build, integrate and reconfigure strategic marketing processes so as to effectively identify and deliver value to international markets (Weerawardena *et al.*, 2015). In a similar vein to Morgan *et al.* (2012), Bucciari *et al.* (2020) emphasized that in the context of new international ventures, DMCs consist of three elements: incremental improvement of current resources, renewal of resources with respect to the requirements of the business environment, and regeneration of capabilities in a way that is adapted to the business environment.

Two streams of research exist regarding DMCs' definition. The first focuses on the consequences of DMCs, such as company performance (e.g. Fang and Zou, 2009; Konwar

et al., 2017). Fang and Zou (2009) conceptualized and operationalized DCs in the marketing function in an empirical study of the impact of DMCs on international joint venture performance and competitive advantage. These authors defined DMCs as “the responsiveness and efficiency of cross-functional business processes for creating and delivering customer value in response to market changes” (p. 744). Similarly, Konwar *et al.* (2017) defined DMCs as an outcome-oriented construct.

The second stream focuses on the process and treats DMCs through the lens of cross-functional processes (Morgan, 2012; Barrales-Molina *et al.*, 2014). Here, DMCs are reconfigured within processes involving different departments of the organization (e.g. Mitrega, 2019; Falasca *et al.*, 2017). Sheng (2017) proposed coordination capabilities, which link diverse resources and are of a character similar to the cross-functional processes. Sheng (2017) points out the need for cooperation among different functional units to ensure effective response to market conditions, as well as better utilization of current market knowledge. For example, Mitrega *et al.* (2019) define DMCs according to the second stream as a “company’s processes that reconfigure its marketing resources for market anticipation” (p. 194). Falasca *et al.* (2017) proposed a definition of DMCs as resource reconfiguration processes intended “to create and deliver customer value in response to market changes” (p. 903). As Mitrega (2019) noted, the first stream of defining DMCs as an outcome-related construct is not aligned clearly with DCV, whereas the second stream is because “the focus of DCV is on processes oriented toward resource transformations not their resource-outcomes, and research on dynamic capabilities in marketing should align with this distinction” (Mitrega, 2019, p. 194).

A broader definition of DMCs was provided by Bucciari *et al.* (2020), who studied the relationship between DMCs and the performance of new international ventures, combining the two definitional streams indicated above. These authors followed Weerawardena *et al.* (2015) and defined DMCs as “the ability to build, integrate, and reconfigure strategic marketing acumen to effectively identify and deliver value to international markets” (p. 3). Similarly, Helfat and Martin (2015), in their study of dynamic managerial capabilities, suggest that it is necessary to measure both aspects, i.e. the process of DCs as well as indirect and direct DCs’ outcomes.

Given the ambiguity in defining IDMCs (Mitrega, 2019), we adopt a definition that includes both indicated streams. In defining IDMCs, we mostly followed the Bucciari *et al.*’s (2020) definition. Specifically, we define IDMCs as *firms’ cross-functional processes for effectively reconfiguring marketing resources in response to environmental dynamism in international markets so as to create and deliver value to international customers*. In the context of our study into transforming countries, we emphasize environmental dynamism (Harris *et al.*, 2009; El Sawy *et al.*, 2010; Pavlou and El Sawy, 2011) as an important driving force in the effective reconfiguration of marketing resources within cross-functional processes (Fang and Zou, 2009).

2.4 Foreign expansion of manufacturing firms from the CEE post-transition economy of Poland as a dynamic research context

The economy of Poland with its own unique characteristics have created a natural dynamically changing environment (Kostova and Hult, 2016), making it an especially suitable setting for studying the effectiveness of firms’ DCs. Broadly speaking, the specifics of CEE economies make them different from other emerging and developed economies. As compared to other emerging economies, the CEE economies have been characterized by more rapid economic growth and institutional development, creating a relatively more unstable and dynamic institutional context for firms’ international growth (Kostova and Hult, 2016). Compared to many other CEE economies, Poland has been better able to more rapidly achieve

high institutional development, although remaining weaker in terms of factor conditions and infrastructural development (Hoskisson *et al.*, 2013). At the same time, companies from Poland and other CEE economies are considered disadvantaged vis-à-vis those from developed economies due to their lack of ownership advantages, relative resource scarcity and weaker institutional support (Caputo *et al.*, 2016). Since 1990, these characteristics have required companies to constantly reconfigure their resources in order to sustain their international competitiveness.

International trade remains important for Poland's economy, with a share of GDP of 105% in 2020 (importing and exporting combined relative to GDP value) (World Bank, 2021). Moreover, exporting contributed to 60% of Poland's average GDP growth between 1995 and 2014 (Hagemejer and Mućk, 2019). The total number of internationally engaged Polish companies (importers and exporters) can be estimated at around 250,000 to 300,000, which accounted for 12.5% of all business entities active in 2016 (around 2 mn) (Cieślík, 2019). Exporting remained the most popular form of foreign market servicing for actively internationalized companies registered in Poland in 2016 (87.5%) (Cieślík, 2019). Among the companies registered in Poland with a majority of Polish capital, 28.4% were exporters, out of which more than 65% exported manufacturing goods in 2016 (Cieślík, 2019). The value of exports from Poland increased by almost ten times between the 1990s and 2020, with a sharp increase after joining the European Union in 2004 (GUS, 2021). Exports by large enterprises (≥ 250 employees) made up 67%, while that of SMEs (10–249 employees) constituted 27% of the 2018 total value of exports (Eurostat, 2021). In 2018, among the firms exporting from Poland, 70% were micro enterprises (0–9 employees), 27.5% were SMEs, while 2.1% were large enterprises (Eurostat, 2021). According to Polish company census data, 74.8% of exports from Poland targeted the European Union market in 2021, with Germany being the major trading partner (28.7% of total exports in 2021) (GUS, 2022). Exports of products from Poland in 2020 exceeded USD 230 bn, while services exported (sold abroad) accounted for more than USD 66 bn (GUS, 2021). In 2018, fully Polish-owned companies dominated in the export of products (55.9%), being more active in terms of foreign trade turnover than companies with foreign ownership (Chojna, 2019). The structure of exported products from Poland in 2019 involved highly processed goods (41%), semi-processed goods (40%) and minerals (18%) (KUKE, 2021). The share of medium-high and high technologically advanced products accounted for 54.4% of the total value of exports, with 44.1% accountable to foreign-owned companies (medium high-tech) and 10.3% to domestic (fully Polish-owned) companies in 2018, though with a sharp increase in the share of the latter after 2008 (Chojna, 2019).

The international growth path of companies originating from Poland resembles a pattern that latecomer firms (LCFs) from East Asia underwent (Mathews, 2002), though with some differences. LCFs' initial disadvantages stemmed from their dislocation from the main international sources of technology, poor access to mainstream international markets, and resource scarcity (Mathews, 2002). Historically, these firms have focused on overcoming their resource position barriers through the iterative process of resource leverage. In doing so, they began by seeking linkages with companies from developed economies through subcontracting arrangements using their cost advantages, or through licencing, which enabled LFCs to learn and imitate foreign solutions (Mathews, 2002). The following stages of international growth are associated with firms strengthening their own position and market power in international value chains, which translates into a gradual increase in markups.

The processes of globalization manifested in the widespread usage of IT technologies and removal of trade barriers, led to fragmentation of global value chains (GVCs) characterized by vertical specialization of activities (Baldwin, 2013). This shift in the nature of GVCs, accompanied by inflows of foreign capital to CEE economies, enabled a structural change in the

exports of these countries (Grela *et al.*, 2017). Consequently, a significant share of CEE economies' exports has been generated by manufacturers located in the middle of GVCs, serving as subcontractors for companies located in developed, mostly EU-based, economies supplying intermediate goods (Hagemejer and Mućk, 2019). Consequently, a major feature of CEE-based firms has been their association with export specialization in capital-intensive goods that has gradually increased value added per unit of output, and therefore the markups generated by these exports (Hagemejer and Mućk, 2019). Against this background, Poland's export-led growth has relied more heavily on production of consumer (vs investment) and final (vs intermediate) goods than the majority of CEE economies (Hagemejer and Mućk, 2019). Indeed, empirical research shows that as manufacturing companies from Poland upgrade their marketing capabilities, they gradually improve their position in both international industrial networks (Kowalik *et al.*, 2020) and international value chains – from being a subcontractor mainly for Western companies, to becoming a firm selling final products on foreign markets under their own brands (Ciszewska-Mlinaric *et al.*, 2020; Mitreġa *et al.*, 2021).

2.5 The COVID-19 pandemic as an example of extreme environmental dynamism

The economic consequences of the COVID-19 pandemic are considered to have met the criteria of an environmental jolt. Environmental jolts are “transient perturbations whose occurrences are difficult to foresee and whose impacts on organizations are disruptive and potentially inimical” (Meyer, 1982, p. 515). As research shows, the consequences of environmental jolts are usually adverse for firms. For instance, Chakrabarti (2015) notes that the economic consequences of a jolt include a sudden and significant drop in demand, leading directly to a decrease in a firm's performance. Environmental jolts occur suddenly, creating discontinuous environmental change with effects difficult to predict. Some prior examples of the studied effect of environmental jolts on organizations are strikes, market deregulation, terrorist attacks, natural disasters or war conflicts (Wenzel *et al.*, 2020). They disrupt established patterns of market structures and inter-firm relationships, and increase the distance between firms and their environment, i.e. lead to a misfit between the organization and its environment (Meyer *et al.*, 1990), which in turn exerts pressure on firms to adapt, i.e. to regain fit (Donaldson, 1987). A direct outcome of environmental jolts is an extreme level of environmental dynamism, which refers to the unpredictability, and thus uncertainty, resulting from the intensity and frequency of changes in the external environment (volatility) (Miller and Friesen, 1983). During such environmental conditions, firms seek to upgrade their resource base through higher-order DCs that are more transformational in nature (Ambrosini *et al.*, 2009). The effectiveness of these resource reconfigurations in response to external dynamism is considered to be lower for established and large firms, as opposed to their small entrepreneurial counterparts, especially high-tech ventures (Schilke, 2014). This is explained by the fact that large firms face numerous difficulties in overcoming organizational inertia due to high organizational complexity, routines and managerial cognitive maps (Gilbert, 2005; Hannan and Freeman, 1984; Tripsas and Gavetti, 2000). Consequently, large firms exhibit a greater preference for lack of or incremental, rather than radical resource reconfigurations (Helfat and Winter, 2011; Schilke, 2014). For instance, Colombo *et al.* (2021), in their study of 340 Italian high-tech entrepreneurial ventures, found that higher levels of new product development (NPD) DCs and internationalization DCs were effective responses during the 2008–2010 global economic crisis in terms of sales growth, but this positive effect in the case of NPD DCs decreased with the size of the enterprise. Moreover, they found a positive effect of the increased use of internationalization DCs on the growth of manufacturing ventures, but not that of ventures in service industries. Conversely, the effect of NPD DCs was positive and significant only for service companies, but not for manufacturing.

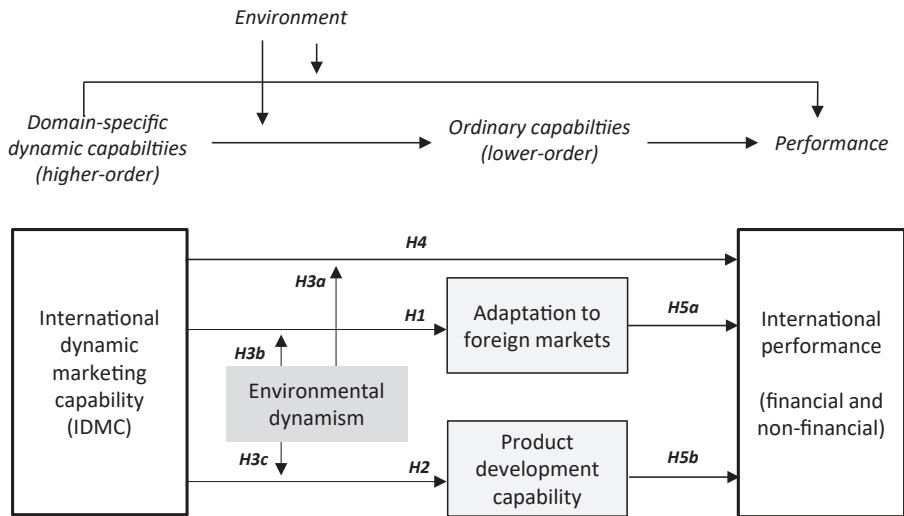
The COVID-19 pandemic is a natural context for testing theory, that is the effectiveness of DCs under conditions of extreme environmental dynamism. In analysing the consequences of the pandemic for Polish-owned export manufacturers, two key issues need closer consideration. The first refers to foreign sales. One of the adverse implications of the coronavirus pandemic was a fall in demand in major export markets for Polish exporters, mainly the German market in the first half of 2020 (Destatis, 2021). According to analysis conducted by the Polish Economic Institute (a public economic think tank), although overall exports from Poland grew by 0.7% in 2020 compared to 2019, the pandemic resulted in a differentiated impact on the exports of firms operating in different industries in Poland (Ambroziak *et al.*, 2021). In contrast to negative foreign sales in services, mostly in HoReCa (hotels, restaurants, catering), exports in business-to-consumer markets (B2C), such as consumer electronics and food products, saw increased sales. However, the majority of Polish-owned export manufacturers operating in business-to-business industries (e.g. subcontractors for foreign car manufacturers, tools and machinery, metal parts, chemical products) experienced a significant deterioration of foreign sales in the first half of 2020 (Ambroziak *et al.*, 2021). The second issue refers to the supply side of exporters registered in Poland in terms of their position in GVCs. Namely, companies' operations were affected, at least in the first half of 2020, by disruptions to foreign supply (see Obłój *et al.*, 2022). Consequently, firms in Poland focused on cost control and were unwilling to engage in risky endeavours (CMMSG, 2020; EY, 2022). One of the important related measures undertaken by Polish companies was redesigning their business relationships (e.g. Obłój *et al.*, 2022). For instance, in their study of 100 Polish SMEs from transportation, freight and logistics (TFL) industry, Mitreġa and Choi (2021) found that dynamic relational capabilities which were helpful in dealing with the pandemic crisis contributed to protecting firms' market position, i.e. through redesigning communication, contractual safeguards, relationship investments and handling of conflicts.

Taken together, we believe that the foreign expansion of manufacturing firms from the post-transition economy of Poland, combined with high environmental dynamism, create an interesting context to examine prevailing international marketing theory and the (dynamic) marketing capabilities view, therefore serving as a foundation for further refinement of international marketing research in non-Western contexts. Moreover, such a unique research setting allows for a better understanding of the boundary conditions for the effectiveness of DCs (Verona and Zollo, 2011).

3. Conceptual model and hypotheses development

In response to limitations in the understanding of how DCs affect performance (see Schilke *et al.*, 2018), this study employs the framework of a two-step process in the link between domain-specific DCs (IDMCs) and performance, and investigates the intermediating factors (lower-order capabilities) by which firms can realize the potential value of their IDMCs. The conceptual model framework (Figure 1) suggests that financial and non-financial international performance are driven by IDMCs through the mediating effect of adaptation to foreign markets and product development capability. We focus on these two factors, as prior research in an international context emphasized the firm's ability to integrate, build and reconfigure resources related to consistent development of new products in multiple geographical markets, and adapt them accordingly (e.g. Sapienza *et al.*, 2006). In addition, the model examines the moderating role of extreme environmental dynamism on the above-mentioned relationship. Thus, whereas lower-order marketing capabilities constitute the VRIN (valuable, rare, inimitable, non-substitutable) resources of firms, IDMCs describe what activities these firms conduct so that their lower-order marketing capabilities are leveraged to improve financial and non-financial performance (Helfat *et al.*, 2007). Moreover, in line with DCV, we propose that the strength of IDMCs will depend on the level of environmental dynamism (e.g. Schilke, 2014).

Figure 1.
Conceptual model



Source(s): Authors own creation

3.1 The mediating effects of adaptation to foreign markets and product development capability

Adaptation versus standardization of international marketing programmes is one of the most widely studied topics in international marketing literature. Numerous studies consider whether and to what extent a marketing strategy should be standardized, or adapted to foreign markets (Fuchs and Köstner, 2015; Larimo and Kontkanen, 2010; Ryans *et al.*, 2003). Some researchers provide arguments in favour of standardization, especially when the market is seen as increasingly homogeneous, meaning that buyers have similar needs and requirements across different foreign markets. On the other hand, many other researchers provide evidence in favour of marketing mix adaptation, stressing that standardization of the marketing mix cannot meet the uniqueness of different foreign markets in terms of buyers' needs (Akgün *et al.*, 2014). Scholars discuss the antecedents and effects of the degree and scope of standardization and adaptation of marketing mix elements (product, price, promotion and distribution) (Sousa and Bradley, 2005; Sousa and Lengler, 2009; Gnizy *et al.*, 2014; Larimo and Kontkanen, 2010; Lu *et al.*, 2010; Fuchs and Köstner, 2015).

For instance, in their empirical study Sousa and Bradley (2005) explored the impact of psychic distance on degree of adaptation of marketing mix elements, and found a positive relationship between all the constructs. A very similar approach to defining marketing programme adaptation to foreign markets was used by Fuchs and Köstner (2015). These authors studied the impact of internal (export market experience) and external (competitive intensity) factors on Austrian SMEs' export success. They found that adaptation of the marketing mix has a positive effect on sales growth and profitability in familiar European and non-familiar non-European and CEE markets. In another study by Lu *et al.* (2010), the adaptive capability category was employed as a higher-order capability. For example, one operationalization of this category was 'Adaptive to respond quickly to a change in product prices' (Lu *et al.*, 2010, p. 430).

The literature also identifies the antecedents of effective marketing programme adaptation (Hultman *et al.*, 2011) and discusses the most beneficial combinations of adaptation and standardization of marketing programmes (Wei and Yazdanifard, 2014),

taking into account differences in firm size, among other factors. For example, for SMEs, orientation towards adaptation of marketing programmes to foreign markets is recommended (Gnizy *et al.*, 2014; Schilke *et al.*, 2009), while large firms (including MNCs), with their ability to achieve economies of scale and scope, should balance adaptation and standardization of marketing programmes (e.g. Vrontis *et al.*, 2009), since full adaptation may be ineffective for them (Schilke *et al.*, 2009). In general, the prevailing view is that the effectiveness of adaptation vs standardization is contingent upon several factors. In other words, a sole focus on the standardization of marketing programmes does not always yield better results than adaptation (Schilke *et al.*, 2009). In this vein, Dow (2006) emphasizes that the benefits of adaptation of marketing programmes are related to the creation of unique value for customers.

The commonly accepted view is that DCs are higher order capabilities and that their impact on performance should be investigated through lower-order firm factors such as resources, activities and processes (Eisenhardt and Martin, 2000). Following this logic, in their empirical study Gnizy *et al.* (2014) found a positive relationship between a proactive learning culture and marketing programme adaptation. A proactive learning culture is understood as an organizational resource upstream of firms' specific capabilities. Consequently, domain-specific DCs may also be related to the category of adaptation to foreign markets. We therefore postulate that IDMCs, as a domain-specific type of DCs, influence the lower order firm's behaviour, i.e. marketing programme adaptation to foreign markets. One IDMCs operationalization (Weerawardena *et al.*, 2015) indicates that a company can combine marketing resources and marketing skills to effectively meet the value expectations of target customers in foreign markets (Weerawardena *et al.*, 2015). The company can thus adapt the content, media and form of promotional activities in foreign markets. Investing more and more in branding in foreign markets (Mitrega, 2019) – which reflects the seizing dimension of IDMCs – is related to a locally integrated foreign brand strategy (Kipnis *et al.*, 2012), which reflects the implementation of adaptation of the marketing mix to the foreign market. We recognize the scarcity of studies on the relationship between IDMCs and marketing programme adaptation to foreign markets. Thus, we hypothesize that.

H1. IDMCs are positively related to adaptation to foreign markets.

Product development capability can be understood as a specialized marketing capability, and as an important factor in firm performance (Morgan *et al.*, 2012; Helm *et al.*, 2020; Möller and Parvinen, 2015; Swan and Luchs, 2011). As a theoretical construct, it refers to activities and processes related to tangible product development stages, such as determining customer needs, identifying ideas for new products, designing protocols for new products, and launching new products (Day, 1994). Two views on product development capability have emerged. The literature associated with the first view indicates that product development capability is a sub-element of dynamic marketing capability (e.g. Fang and Zou, 2009; Morgan, 2012; Morgan *et al.*, 2012; Hoque *et al.*, 2022). In the context of international joint-ventures (IJVs), Fang and Zou (2009), in conceptualizing DMCs, identified product development as one of three components, alongside customer relationship management and supply chain management. In their view, DMCs are a variation of DCs representing a higher-order capability, while product development consisting of detailed activities represents a lower-order capability. The second stream of research conceptualizes product development capability as a separate construct that relates to a somewhat narrower understanding of DMCs, viewed in terms of knowledge acquisition and development (Subramaniam and Venkatraman, 2001; Zhang and Zhu, 2020). Morgan *et al.* (2012) distinguished two groups of DMCs: (1) architectural, related generally to information management, and (2) specialized, related to the implementation of various marketing activities, including product development capability. In their study, Helm *et al.* (2020) analyse

the relationship between marketing intelligence capability (referring to Morgan *et al.*'s architectural capability) and product development capability (specialized marketing capability). The positive link between capabilities related to marketing knowledge and product development or new product development capability has long been accepted knowledge (e.g. Subramaniam and Venkatraman, 2001). For instance, a recent study by Zhang and Zhu (2020) found that the acquisition of market knowledge has a greater impact on the speed of new product development than technological knowledge.

Recently, Hoque *et al.* (2022), in their empirical study of Bangladeshi exporters representing eight industries, conceptualized IDMCs as a higher-order latent construct that includes four specific lower-order marketing capabilities, that is proactive market orientation, brand management capability, new product development capability and customer relationship management capability. Similarly, a study by Zadykiewicz *et al.* (2020) investigated the relationship between lower-order capabilities, such as proactive customer orientation and joint learning capabilities, and the collaborative development of innovative tangible products and technology in the automotive industry in a relationship between a weaker supplier and a stronger buyer (e.g. Makkonen *et al.*, 2021; Siemieniako *et al.*, 2022; Siemieniako and Kaliszewski, 2022).

Research to date has focused on identifying IDMCs as enhancing product development capability – a key success factor for exporters competing in highly demanding sectors (Lages *et al.*, 2009). Sheng (2017) conceptualizes IDMCs in terms of the tight coordination of functional units to ensure better use of the company's foreign market knowledge. This high level of cooperation among functional units, including sales and R&D involvement in product development, helps ensure an effective response to market conditions (Sheng, 2017). One such company response can be to improve efficiency in product development and new product launches in foreign markets (Mitrega, 2019).

However, the literature has conceptualized IDMCs to date solely in relation to marketing knowledge. However, marketing activities in foreign markets involve great uncertainty and the need to adapt marketing resources and skills to meet the demand in foreign markets (Weerawardena *et al.*, 2015), which justifies treating IDMCs as marketing activities that are not merely limited to promotion and branding, but also include related activities, such as those associated with product development, i.e. which require cross-functional collaboration (see section 4.2 and Table 2).

Consequently, following Mitrega (2019) we argue that IDMCs, as a higher-order construct, include marketing-related processes and activities that lead to, but are separate from tangible product development. Therefore, similarly to Cao *et al.* (2019), we propose that product development capability, as a lower-order construct (see section 4.2 and Table 2), is developed by the deployment of IDMCs and allows higher performance to be achieved in foreign markets. Therefore, we formulate the following hypothesis.

H2. IDMCs are positively related to product development capability.

3.2 Relationship between IDMCs and international performance with and without mediating effects

As stated earlier, the main feature of DCs is the firm's ability to respond effectively to external market changes (Teece *et al.*, 1997) in order to create and deliver value to customers in foreign markets, which is an important factor in financial and non-financial performance (e.g. Day, 1994). Several studies found that DCs do not necessarily by themselves lead to improved performance, but rather it is the resource configurations created by DCs that lead to the final outcome (Eisenhardt and Martin, 2000), contingent on environmental dynamism (Wilden *et al.*, 2013). In general, most studies show that DCs positively impact firms' international performance (e.g. Drnevich and Kriauciunas, 2011; Wójcik and Ciszewska-Mlinarič, 2021;

Khan *et al.*, 2020). This is related to the broader view that export marketing strategy is positively related to export performance (Katsikeas *et al.*, 2000; Leonidou *et al.*, 2002).

The theoretical discussion covers two streams, the first linking IDMCs directly to a firm's international performance, and the second indicating the presence of a mediating effect in that relationship. The ambiguity of defining of IDMCs (Mitrega, 2019) motivated us to draw from Buccieri *et al.*'s (2020) study. These authors embrace both streams in their research into high-technology international new ventures from India (regarding the direct and indirect effect of IDMCs on performance).

According to Helfat and Martin (2015), measuring the performance of DCs should first address the impact of DCs on intermediate outcomes, such as lower-order capabilities, followed by examination of this impact on firm performance. In general, research on the mechanisms mediating the relationship between DCs and performance is rather limited. In their review, Schilke *et al.* (2018) pointed to a general mechanism mediating a company's DCs and performance, which is the company's portfolio of resources and capabilities (e.g. Fainshmidt *et al.*, 2016; Pavlou and El Sawy, 2011). For example, in research by Pavlou and El Sawy (2011), operational capabilities mediate the relationship between DCs and performance in new product development.

Furthermore, research on the mechanisms mediating the relationship between IDMCs and international performance is scarce. Interestingly, Morgan *et al.* (2018) found in their review that the mediators (mechanisms) of the relationship between marketing capabilities and company performance were used more often in studies of firms operating in international markets than in research on firms focused only on their domestic markets. The identified mediators are: competitive advantages, business strategy, implementation effectiveness and efficiency, innovation, and international-related factors such as international commitment, involvement and foreign market entry modes (Morgan *et al.*, 2018). In general, the understanding of the mediating mechanisms of how IDMCs impact financial and nonfinancial firm performance is not yet mature and remains under-researched.

The literature is ambiguous regarding the impact of adapting a marketing programme to foreign markets on international performance (Fuchs and Köstner, 2015). A number of studies confirm that adaptation to the requirements of foreign markets positively affects firm performance, e.g. with regard to product and price adaptation (e.g. Cavusgil and Zou, 1994; Shoham, 1999; Calantone *et al.*, 2006) or with regard to promotion adaptation (Shoham, 1999). Another group of empirical studies indicates a negative relationship between adaptation to foreign markets and company performance, e.g. with regard to product adaptation (Lages *et al.*, 2008), or in the case of promotion adaptation (Cavusgil and Zou, 1994).

In past studies, it was quite common to use the adaptation of a marketing programme to a foreign market as a mechanism to mediate the relationship between different categories and a company's international performance. For example, Lu *et al.* (2010) used managerial ties and institutional capital, while Fuchs and Köstner (2015) – export market experience and competitive intensity. In contrast, Gnizy *et al.* (2014) used the adaptation of a marketing programme as a mediator in the relationship between a proactive learning culture and foreign launch success. However, studies have not used the category of adaptation to foreign markets as a mediating mechanism in the relationship between IDMCs and company performance.

Similarly, the tangible product development capability category has not yet been used as a mediator in the link between IDMCs and company performance. Yet, Helm *et al.* (2020) categorized product development capability as a specialized marketing capability (Morgan *et al.*, 2012), and used it as a mediating factor in the relationship between marketing intelligence capability, which resonates with Morgan *et al.*'s (2012) architectural capability, and company performance.

Given these arguments, we propose not only a hypothesis regarding the direct relationship between IDMCs and international performance, but also two hypotheses considering the mediating effects within this relationship, as follows.

H3. IDMCs are positively related to a firm's international performance (financial and non-financial).

H4a. The relationship between IDMCs and international performance is mediated by adaptation to foreign markets.

H4b. The relationship between IDMCs and international performance is mediated by product development capability.

3.3 *The moderating effect of environmental dynamism on IDMCs*

Strategic fit theory indicates that a firm should fit its strategy to its environment (Venkatraman, 1989). The key issue is that a firm reacts to "the environment as an exogenous variable and adjusts its marketing strategy and/or organizational form to match the environment" (Katsikeas *et al.*, 2006, p. 869).

There is an abundance of literature on environmental dynamism as one of the major contingency factors in the DC-performance relationship (Fainshmidt *et al.*, 2016; Karna *et al.*, 2016). However, research on the impact of domain-specific DCs on performance is limited. Environmental dynamism refers to the rate and unpredictability of environmental change (Miller and Friesen, 1983; El Sawy *et al.*, 2010; Harris *et al.*, 2009). Some studies refer to environmental turbulence, which can be defined as the frequency and amplitude of change in the environment and overall uncertainty (Duncan, 1972). For instance, Pavlou and El Sawy (2011) identify market and technological turbulence as two primary sources of environmental turbulence, and found that environmental turbulence positively moderates the effect of DCs on operational capabilities related to new product development (NPD), but has a negative moderating (attenuating) role on the effect of operational capabilities on NPD performance. Mitre ga and Pfajfar (2015) used environmental uncertainty as a moderator between business relationship process management (as a domain-specific higher-order capability) and company performance. They found strong and statistically significant support for the tested hypothesis about this relationship. We follow Miller and Friesen (1982) and Bucciari *et al.* (2020) in operationalizing environmental dynamism, which relates to the market dynamism and technological dynamism employed by Pavlou and El Sawy (2011). Previous research indicates that environmental dynamism provides new opportunities (e.g. Sull, 2009; Welter and Smallbone, 2011), and DC literature emphasizes that the probability of resource reconfiguration increases in rapidly changing environments as incentives are created for seeking new ways of doing things, and thus create a need for resource reconfiguration so as to take advantage of new opportunities (Teece *et al.*, 1997). However, there is scarcity of studies on how environmental dynamism affects the relationship between domain-specific DCs and performance.

The category of environmental dynamism is used as an antecedent to IDMCs (e.g. Schilke *et al.*, 2018), but is primarily the most studied moderator in the link between IDMCs and firm performance (Fang and Zou, 2009; Schilke, 2014; Kaleka and Morgan, 2019; Morgan *et al.*, 2018). As Morgan *et al.* (2018) note, the findings of studies are ambiguous when it comes to the moderating effect of environmental dynamism on the relationship between DMC and firm performance. On the one hand, empirical studies provide evidence of a significant and positive moderating effect of high environmental dynamism or turbulence, i.e. strengthening the positive effect of DMCs on performance (e.g. Bucciari *et al.*, 2020; Fang and Zou, 2009). However, others demonstrate the opposite (e.g. Schilke, 2014) or a lack of impact of high volatility (e.g. Wang *et al.*, 2015). For instance, Schilke (2014) showed that a higher level of

environmental dynamism weakens the association between alliance and NPD DCs and firm performance.

Moreover, Yiu *et al.* (2007) found lower adaptation flexibility (the ability to implement different levels of marketing mix adaptation to foreign market requirements) under conditions of a rapidly changing environment. In turn, an empirical study by Alashban *et al.* (2002) showed that in a highly dynamic environment, multinational corporations often fail to adapt their marketing programmes sufficiently.

In terms of adaptation, the recent pandemic posed new threats in terms of a drop in demand, but also generated opportunities for creating value for buyers in international markets, as it changed clients' needs (Colombo *et al.*, 2021). We therefore expect that exporters faced with a drop in demand would likely intensify their adaptation efforts to foreign markets and change the way they develop their products, seeking to improve their competitiveness. However, discontinuous environmental change at the same time significantly increases uncertainty in the short term, manifested through different levels of demand drop across countries, industries and a change in direction of resource and capabilities reconfiguration (Helfat *et al.*, 2007). This claim finds empirical support in a study by Schilke (2014), who finds a lower effectiveness of DCs under conditions of higher levels of dynamism. In this vein, Schilke *et al.* (2018, p. 406) note that this is because "*environmental changes may be too abrupt and unforeseeable to fully leverage planned strategic change*". Consequently, firms may not be able to fully use the potential of exploratory learning and experimentation and thus benefit from the upgrade of new product development capability and suitable adaptation of these products to foreign markets. Accordingly, we expect that the direct linkage between IDMCs and performance, as well as between IDMCs and lower-order categories (adaptation to foreign markets and product development capability), will be less pronounced for export manufacturers operating under conditions of radical environmental shift. We therefore posit the following hypotheses.

- H5a. The positive relationship between IDMCs and international performance is weaker when there is higher environmental dynamism.
- H5b. The positive relationship between IDMCs and adaptation to foreign markets is weaker when there is higher environmental dynamism.
- H5c. The positive relationship between IDMCs and product development capability is weaker when there is higher environmental dynamism.

4. Research methodology

4.1 Data collection

The hypothesized relationships (Figure 1) were tested on cross-sectional data collected in August–September 2021 from a sample of 277 internationally-oriented Polish manufacturing companies. We therefore captured the situation of almost two years of extreme environmental dynamism (the pandemic). The research concept, sampling assumptions and questionnaire were developed by the authors, while the implementation of the study was entrusted to an external professional research agency (CBM Indicator) with expertise in projects for academia. The sampling operator was the BISNODE database, the largest database of enterprises operating in Poland. We focused only on the manufacturing industry. To increase data quality, we decided on computer assisted telephone interviewing (CATI), with interviewers trained in instrument questions. The interviews began with a set of questions to determine if a company – despite the respondent's initial consent to the interview – could be included in the final sample.

Taking into account the research objective, we focused on manufacturing companies that met following detailed selection criteria. First, we focused only on companies for which internationalization was an important strategic direction, i.e. the foreign sales to total sales ratio

in 2020 was at least at the level of 10% or higher. Second, company ownership was taken into account as a selection criterion. Only companies with a majority share of Polish capital (at least 51%) were included in the final sample. In this way, our sample included companies with local/Polish decision-making autonomy, excluding from the sample enterprises registered in Poland, but which were subsidiaries of multinational corporations or with a dominant share of foreign capital. Third, only companies with at least one full year of operation (i.e. founded no later than in 2019) and those employing at least 10 people at the end of 2020 were considered for the sample. Fourth, we verified the interviewee's position (CEO, owner, sales executive director, senior level export director) and experience in the company (at least one year in the position), and if an interviewee did not meet the above criteria, the interview was ended. We targeted owners and senior level managers as key informants in order to ensure that the participants had authority to make key decisions related to foreign markets (Myers, 2009).

The method employed for the research, i.e. one informant per company, and one measurement instrument, can potentially lead to common method bias/common method variance (CMV) (Podsakoff *et al.*, 2003; Chang *et al.*, 2010). Thus, to address this concern, we applied *ex ante* measures: the questionnaire was pretested to eliminate ambiguity, and informants were assured about the confidentiality of their responses and that the results for individual companies would not be disclosed. As an *ex-post* remedy, we used a check for respondent competency (Morgan *et al.*, 2004) to ensure that the interviewers identified appropriate informants. After the completion of the interview, the interviewees were asked to evaluate three statements concerning: (1) their impact on key decisions related to the international activities of their companies; (2) their engagement in maintaining relationships with key foreign partners; and (3) their confidence in answering the questions. We excluded from further analysis three questionnaires that indicated inadequate levels of informant competency, with an average score on a seven-point scale of 4 or below. In the final sample ($n = 277$), the average score was 6.1, indicating an appropriate level of respondents' competency for them to be able to answer the survey questions. Finally, a post hoc Harman single-factor test was employed to detect CMV. The test results indicated that the first factor accounted for 32.51% of the total variance, with no single factor emerging as dominant; therefore, CMV was deemed not be a problem in the present study (Podsakoff *et al.*, 2003). Details on the characteristics of the sample are presented in Table 1.

4.2 Measures

In order to secure reliability and validity, the scales used in our study were adapted from prior studies, with only a few relevant adjustments where necessary. As the questions were originally in English, we developed a Polish version of the survey instrument, following the translation-back-translation procedure. The operationalization of all constructs, including the items and their respective factor loadings, as well as the results of the reliability analysis, are presented in Table 2.

4.2.1 Financial and non-financial international performance. In line with the call by Hult *et al.* (2008) to apply multiple performance measures in international business (IB) studies, we differentiated between financial and non-financial performance. Such an approach has also been recently adopted in marketing research; for instance, when studying the relationship between dynamic marketing capabilities and INV performance, Bucciari *et al.* (2020) differentiated between financial and market performance. The measures for performance in foreign markets were derived from IB literature (Evans *et al.*, 2008; Azar and Drogendijk, 2014). Respondents were asked to rate the company's financial indicators in international markets in 2020 in comparison to 2019 (during the COVID-19 pandemic) on a seven-point scale, where 1 meant "a very significant decrease (by over 20%)", 2 – "a decrease (by 11–20%)", 3 – "a slight decrease (up to 10%)", 4 – "no change", 5 – "a slight increase (up to 10%)", 6 – "an increase (by 11–20%)";

Sample characteristics	Frequency	%		Freq	%
Firm age (years of operation)			Position		
5–10 years	32	12	CEO/owner	20	7.2
11–20 years	54	19	CEO	47	17
21–30 years	59	21	Sales executive director	114	41.2
>30 years	132	48	Senior level directors (exports/foreign sales)	96	34.7
Firm size (number of employees)			Gender		
10 to 249	158	57	Male	198	71.5
250 to 499	85	31	Female	79	28.5
500 and more	34	12	Other	0	
Ownership			Experience at company		
100% of Polish capital	221	79.8	2–5 years	68	25
Majority share of Polish capital (75–99%)	27	9.7	6–10 years	96	35
Majority share of Polish capital (51–74%)	29	10.5	11–15 years	64	23
			Above 15 years	49	18
Industry			Degree of internationalization (FS/TS)		
High- and medium-high tech	138	50	10 to 25	109	39
Medium-low and low-tech	139	50	26 to 50	101	36
Sector/segments			51 to 75	47	17
Only B2B	194	70	76 to 100	20	7
Only B2C	11	4			
B2B and B2C	72	26	Internationalization scope (no. of markets)		
Value chain position			2 to 5	32	12
Firm sells semi-finished products, further processed by buyers on foreign markets or finished products sold under the buyers' brand (a subcontractor for buyers on foreign markets)	105	38	6 to 10	86	31
			11 to 20	114	41
			>20	45	16
			Internationalization speed (years from founding)		
			3 and less	117	42
			4 to 10	129	47
Firm sells finished products on foreign markets under its own brand (products with the firm's brand are used by end customers)	172	62	Above 10	31	11
			Firm international experience (no. of years selling abroad)		
			Up to 10	46	17
			11 to 20	81	29
			21 to 30	71	26
			>30	79	29

Source(s): Authors' own creation

Table 1.
Descriptive statistics of
the sample
companies (N = 277)

and 7 – “a very significant increase (by more than 20%)”. To operationalize non-financial performance in international markets, we used six items. Respondents were asked to indicate the level of their company's international performance in 2020 compared to 2019 on a scale of 1–7, where 1 corresponded to “very unsatisfactory” and 7 – “very satisfactory” in the following areas: (1) increase in market share; (2) achievement of strategic goals in foreign markets; (3) marketing effectiveness; (4) effectiveness of distribution; (5) company reputation; (6) overall satisfaction with performance in foreign markets.

Construct and measures	Factor loading	CITC	α
<i>IDMC (CR = 0.95, AVE = 0.70)</i>			
1 We check if our marketing tools applied in foreign markets are effective	0.780	0.815	0.956
2 We constantly search for gaps in our marketing capabilities used in foreign markets	0.796	0.815	
3 We check how our marketing methods applied in foreign markets compare to methods used by competitors in foreign markets.	0.892	0.803	
4 We invest more and more in branding in foreign markets	0.885	0.834	
5 The activities of functional units (marketing, sales, R&D, production) are tightly coordinated to ensure better use of our foreign market knowledge.	0.868	0.865	
6 There is a high level of cooperation among functional units in setting our firm's goals and priorities concerning foreign markets to ensure an effective response to market conditions.	0.824	0.830	
7 We modify our image in foreign markets in response to situation in the environment	0.810	0.803	
8 We introduce conscious changes with regard to the desired customer brand associations in foreign markets.	0.872	0.841	
9 We combine marketing resources and skills to effectively meet the value expectations of targeted customers in foreign markets.	0.765	0.820	
<i>Adaptation to foreign markets (CR = 0.82, AVE = 0.54)</i>			
1 Product in terms of quality, design, warranties, labelling and brand name	0.653	0.559	0.803
2 Price in terms of price discount policy, margins, credit concession and payment security	0.911	0.679	
3 Promotion in terms of advertising theme/message, advertising and promotion content, advertising media strategy, sales promotion tools, and advertising and promotion budget.	0.557	0.591	
4 Distribution in terms of channels of distribution, control over distribution channels, transportation strategy and budget for distribution.	0.762	0.677	
<i>Product development capability (CR = 0.82, AVE = 0.59)</i>			
1 Our company has skills in developing new products	0.880	0.761	0.860
2 Our company has ability to upgrade existing products	0.802	0.649	
3 Our company has ability to transform R&D results to products	0.876	0.750	
4 Our company is often the first to market with new products and services	0.647	0.693	
5 Our new product introduction has increased over the last 3 years	0.586	0.630	
<i>Environmental dynamism (CR = 0.76, AVE = 0.69)</i>			
1 In our industry, methods of production change often and in major ways	0.887	0.820	0.874
2 Our firm must change its marketing practices frequently	0.761	0.658	
3 In our environment, new business models evolve frequently	0.837	0.809	
<i>Financial performance on foreign markets (CR = 0.74, AVE = 0.66)</i>			
1 Sales volume/level in foreign markets	0.782	0.786	0.839
2 Profitability of foreign sales	0.825	0.731	
3 Value of sales from new products/services on foreign markets	0.826	0.612	
<i>Non-financial performance on foreign markets (CR = 0.92, AVE = 0.67)</i>			
1 Foreign markets' share growth	0.739	0.851	0.934
2 Achievement of foreign markets' strategic goals	0.834	0.896	
3 Marketing effectiveness on foreign markets	0.876	0.882	
4 Effectiveness of distribution in foreign markets	0.837	0.786	
5 Firm's reputation in the foreign markets	0.738	0.549	
6 Overall satisfaction with the performance in foreign markets	0.879	0.879	

Note(s): Construct reliability: alphas for all above 0.7, CITC – corrected item-total correlation above 0.3; Composite reliability (CR) above 0.6

Convergent validity: Average factor loading for all components above 0.7; Discriminant validity: average variance extracted (AVE) > square of correlations for all pairs of constructs; AVE above 0.5

Source(s): Authors' own creation

Table 2.
Construct measures
and reliability index

International dynamic marketing capability was operationalized by 9 items capturing the three major categories of DCs: sensing, seizing and reconfiguring (Teece, 2007). Since at the time of designing the questionnaire there was no validated scale specific for IDMCs, we decided to adapt the scale developed by Mitrega (2019) for DMCs by adding the international market context to the original items. We also decided to enrich the scale by including two items derived from Sheng (2017) reflecting the seizing aspect in the operationalization of IDMCs, i.e. “The activities of functional units (marketing, sales, R&D, production) are tightly coordinated to ensure better use of our foreign market knowledge”; “There is a high level of cooperation among functional units in setting our firm’s goals and priorities concerning foreign markets to ensure an effective response to market conditions” (no. 5 and 6 in Table 2), and one item strengthening the operationalization of reconfiguration aspect of IDMCs from Weerawardena et al. (2015) i.e. “We combine marketing resources and skills to effectively meet the value expectations of targeted customers in foreign markets” (no. 9 in Table 2). The measures employed used a seven-point Likert scale ranging from “1 – strongly disagree” to “7 – strongly agree”.

In line with Morgan et al.’s (2012) conceptualization of marketing capabilities, *product development capability* represents “specialized marketing capabilities’ that enable firms to act and deploy resources aligned with intended marketing strategy, and facilitates “the adjustments of execution tactics and resource deployments during strategy implementation” (Helm et al., 2020, p. 21). The scale used in our study resonates well with prior operationalizations (e.g. Helm et al., 2020; Morgan et al., 2012) as it reflects organizational skills and abilities in terms of development of new products, upgrading existing products, transforming/exploiting R&D results to products, and launching first-to-market and new-to-company products. The key difference between our and prior operationalizations (Helm et al., 2020) is that we have not included the aspects of after-sale activities. Respondents were asked to rate their firm’s capability relative to their key competitors on each of the five items, using a seven-point scale ranging from “1 – much weaker than the competition” to “7 – much better than the competition”.

To operationalize *environmental dynamism*, which is manifested “by the amount and unpredictability of change in customer tastes, production or service technologies, and the modes of competition in the firm’s principal industries” (Miller and Friesen, 1983, p. 233), we followed Buccieri et al. (2020), who adapted items originally developed by Miller and Friesen (1982, 1983). These measures employed a seven-point Likert scale ranging from “1 – strongly disagree” to “7 – strongly agree”.

4.2.2 Adaptation to foreign market. The measurement of adaptation was derived from prior research (Fuchs and Köstner, 2015; Sousa and Bradley, 2005; Sousa and Lengler, 2009). Managers were asked to indicate the level of adaptation/adjustment to foreign markets with regard to four items of the marketing mix and their extended definitions, using a seven-point scale ranging from “1 – very low level/range of adjustment to foreign markets” to “7 – very high level/range of adjustment”.

4.2.3 Control variables. The study uses six control variables relevant for studies concerning the performance of internationally-oriented firms. First, the speed of firm internationalization (accepted as the firm’s age at the time of achieving the first revenues from foreign markets) points to the phenomenon of “learning advantage of newness” (Autio et al., 2000; Sapienza et al., 2006). Early internationalization in an organizational life cycle may reflect a higher organizational capacity to assimilate new foreign market knowledge, recognize opportunities and rapidly act on them. Second, we also checked for the firm’s international experience measured as the number of years that the firm has been operating in foreign markets, as well as for the firm’s international scope (number of foreign markets). Third, the firm’s size (operationalized by the number of employees) serves as a proxy of the available resources, and needs to be checked in performance studies. For value chain position

(VCP), a binary variable, respondents were asked which statement better suits their company's situation: "Firm sells semi-finished products, further processed by buyers on foreign markets, or finished products sold under the buyers' brand (a subcontractor for buyers on foreign markets)" – coded as "0"; "Firm sells finished products on foreign markets under its own brand (products with the firm's brand are used by end customers)" – coded as "1". Finally, the industry was operationalized with a binary variable, taking the value 1 for high-tech and medium high-tech industries. Correlations and descriptive statistics for all of the variables are shown in Table 3.

5. Analysis and results

We used hierarchical multiple regression analysis to test our hypotheses. The results are shown in Tables 4 and 5. The variance inflation factors (VIF) associated with each regression coefficient did not show significant multicollinearity (VIF <1.83), suggesting no serious problems with multicollinearity. In the hierarchical regression analysis presented in Tables 4 and 5, in the first step we included only control variables, then we added IDMCs in the second step to test H1 and H2 (Table 4) and H3 (Table 5), followed by environmental dynamism in the third step, and interaction terms in the fourth step to test the moderator hypotheses H5b and H5c (Table 4) and H5a (Table 5). In the final steps (Table 5 only) we added the mediators, i.e. adaptation to foreign markets and product development capability, in order to test H4a and H4b.

H1 predicts that IDMCs have a positive effect on adaptation to foreign markets. Model 2 in Table 4 confirms the result ($\beta = 0.63, p < 0.001$). Thus, H1 is supported. H2 predicts that IDMCs have a positive effect on product development capability. Model 6 in Table 4 also confirms the result ($\beta = 0.30, p < 0.001$). Therefore, H2 is also supported.

The results for the moderating effects of environmental dynamism for the relationships between IDMCs and adaptation to foreign markets (H5b), and between IDMCs and product development capability (H5c) are also presented in Table 4. As can be seen, environmental dynamism negatively moderated the effect of IDMCs on adaptation to foreign markets (Model 4: $\beta = 0.18, p < 0.001$), as well as the effect of IDMCs on product development capability (Model 8: $\beta = 0.11, p < 0.001$), supporting respectively H5b and H5c (Figure 2).

Table 5 presents the results of the hierarchical regression analysis for the relationships between IDMCs and international performance, including the moderating and mediating effects as postulated in hypotheses H3, H5a, H4a and H4b. Since we differentiate between financial and non-financial performance, all the models in Table 5 are run separately for these

	1	2	3	4	5	6	7	8	9	10	11	12
1 Financial performance	1											
2 Non-financial performance	0.64**	1										
3 IDMC	0.18**	0.23**	1									
4 Environment dynamism	0.23**	0.10	0.27**	1								
5 Product development capability	0.22**	0.22**	0.32**	0.43**	1							
6 Adaptation to foreign markets	0.23**	0.28**	0.63**	0.22**	0.24**	1						
7 Firm size	0.09	0.09	0.10	0.07	0.08	0.08	1					
8 Firm int. experience	-0.06	-0.06	0.00	-0.02	-0.06	0.01	0.08	1				
9 Firm int. speed	-0.07	-0.01	-0.01	-0.11	0.02	0.05	0.00	0.00	1			
10 Firm int.scope	0.12*	0.08	0.00	-0.26**	-0.03	-0.01	0.28**	-0.11	0.08	1		
11 VCP (Own brand)	0.02	0.02	-0.03	0.03	0.16**	0.01	0.05	-0.04	0.04	0.13*	1	
12 High-/Medium-high tech	0.06	0.02	0.05	0.24**	0.35**	0.05	-0.04	0.06	0.03	0.10	0.10	1

Table 3.
Descriptive statistics
and correlations

Note(s): *p < 0.05; **p < 0.01
Source(s): Authors' own creation

Variables	DV: Adaptation to foreign markets			VIF	DV: Product development capability					
	Model 1	Model 2	Model 3		Model 4	Model 5	Model 6	Model 7	Model 8	VIF
IDMC		0.634***	0.620***	0.595***	1.12		0.299***	0.225***	0.209***	1.12
EnvDyn			0.059	0.022	1.37			0.302***	0.279***	1.37
IDMC x EnvDyn				−0.175***	1.10				−0.108*	1.10
Firm size	0.095	0.019	0.010	0.032	1.17	0.131*	0.095†	0.049	0.062	1.17
Firm_Int_exp	−0.006	0.004	0.009	0.020	1.05	−0.100 †	−0.096†	−0.071	−0.064	1.05
Firm_Int_speed	0.053	0.061	0.067	0.066	1.02	0.015	0.018	0.046	0.045	1.02
Firm_Int_scope	−0.051	−0.030	−0.010	−0.028	1.31	−0.129*	−0.119*	−0.017	−0.028	1.31
VCP (Own brand)	0.004	0.029	0.026	0.013	1.04	0.125*	0.136*	0.123*	0.115*	1.04
High-tech	0.058	0.016	0.000	0.022	1.15	0.366***	0.347***	0.265***	0.278***	1.15
R ²	0.014	0.409	0.412	0.440		0.169	0.256	0.325	0.336	
Adj. R ²	0.008	0.394	0.394	0.421		0.150	0.237	0.305	0.313	
F	0.63	26.11***	23.45***	23.27***		9.14***	13.25***	16.15***	14.99***	
Change in R ²		0.395	0.003	0.028			0.088	0.069	0.011	
F-change		180.01***	1.19	13.23***			31.70***	27.36***	4.24*	

Note(s): Cell entries are standardized regression coefficients. †p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001

Source(s): Authors' own creation

Note(s): Cell entries are standardized regression coefficients. †p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001

Source(s): Authors' own creation

Table 4. Hierarchical regression results for adaptation to foreign market and product development capabilities as dependent variables (DV)

Table 5.
Hierarchical regression
results for financial
and non-financial
performance as
dependent
variables (DV)

	DV: Financial performance					DV: Non-financial performance										
	Model 1a	Model 2a	Model 3a	Model 4a	Model 5a	Model 6a	Model 7a	VIF	Model 1b	Model 2b	Model 3b	Model 4b	Model 5b	Model 6b	Model 7b	VIF
IDMC		0.175**	0.116 [†]	0.117 [†]	0.001	0.088	-0.030	1.82		0.230***	0.215***	0.211***	0.082	0.175**	0.043	1.82
EnvDyn			0.241***	0.242***	0.237***	0.204**	0.199**	1.49			0.058	0.051	0.046	0.003	-0.003	1.49
IDMC x				0.003	0.037	0.017	0.052	1.17				-0.033	0.005	-0.014	0.025	1.17
EnvDyn					0.195*		0.197*	1.79					0.216**		0.220**	1.79
Adaptation_																
For_Mrk																
Prod_Dev_						0.135 [†]	0.138*	1.51						0.171*	0.175*	1.51
Cap						(p = 0.058)										
Firm size	0.076	0.055	0.018	0.018	0.012	0.009	0.003	1.18	0.080	0.052	0.044	0.048	0.041	0.037	0.030	1.18
Firm_Int_	-0.058	-0.055	-0.035	-0.035	-0.039	-0.027	-0.030	1.06	-0.061	-0.057	-0.052	-0.050	-0.054	-0.039	-0.043	1.06
exp																
Firm_Int_	-0.079	-0.077	-0.055	-0.055	-0.068	-0.061	-0.074	1.03	-0.013	-0.010	-0.004	-0.005	-0.019	-0.012	-0.027	1.03
speed																
Firm_Int_	0.093	0.099	0.180**	0.181**	0.186**	0.184**	0.190**	1.31	0.053	0.060	0.080	0.077	0.083	0.082	0.088	1.31
scope																
VCP (Own	0.001	0.008	-0.003	-0.003	-0.006	-0.018	-0.022	1.06	0.009	0.018	0.016	0.013	0.010	-0.006	-0.010	1.06
brand)																
High-tech	0.058	0.047	-0.019	-0.019	-0.024	-0.057	-0.062	1.27	0.026	0.011	-0.005	-0.001	-0.006	-0.049	-0.055	1.27
R ²	0.031	0.061	0.105	0.105	0.126	0.117	0.139		0.016	0.068	0.070	0.071	0.098	0.091	0.118	
Adj. R ²	0.009	0.036	0.078	0.075	0.093	0.084	0.103		0.006	0.044	0.043	0.040	0.064	0.057	0.081	
F	1.42	2.49*	3.95***	3.48***	3.84***	3.52***	3.88***		0.74	2.80**	2.54*	2.28*	2.88**	2.66**	3.22***	
Change in		0.030	0.044	0.00	0.021	0.012	0.013			0.052	0.003	0.001	0.026	0.020	0.020	
R ²																
F-change		8.66**	13.18***	0.00	6.47**	3.63 [†]	3.88*			14.94***	0.73	0.28	7.74**	5.71**	6.10*	

Note(s): Cell entries are standardized regression coefficients. [†]p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001

Source(s): Authors' own creation

Note(s): Cell entries are standardized regression coefficients. [†]p < 0.10; *p < 0.05; **p < 0.01; ***p < 0.001

Source(s): Authors' own creation

two dependent variables. [H3](#) predicts the direct positive effect of IDMCs on a firm's international performance (financial and non-financial). Models 2a and 2b in [Table 5](#) confirm the result for both financial performance ($\beta = 0.18, p < 0.01$) and non-financial performance ($\beta = 0.23, p < 0.001$). Thus, [H3](#) is supported. [H5a](#) predicts that the IDMC-performance relationship is moderated by environmental dynamism. However, the moderating effect of environmental dynamism was neither confirmed for the IDMC-financial performance relationship (Model 4a: $\beta = 0.003, p > 0.1$) nor for the IDMC-non-financial performance relationship (Model 4b: $\beta = -0.033, p > 0.1$), thus [H5a](#) was not supported.

To test whether adaptation to foreign markets and product development capability are mediators for the effects of IDMCs on performance, we followed the suggestion of [Baron and Kenny \(1986\)](#). The results presented above indicate that the first two conditions for the mediating effects were met (i.e. IDMCs affect performance; IDMCs affect adaptation and product development capability). Thus, the mediating variables must have an effect on performance (as a dependent variable), and finally, we need to evaluate whether the significance of the direct relationships between IDMCs and performance is reduced after the inclusion of the mediating variables. To examine the latter, we included the mediators in the regression model along with IDMCs.

As indicated in Model 5a and 5b in [Table 5](#), after the inclusion of adaptation to foreign markets as a predictor of performance, IDMCs became insignificant as a predictor of financial performance (Model 5a: $\beta = 0.001, p > 0.1$) and non-financial performance (Model 5b: $\beta = 0.082, p > 0.1$), while the direct effect of adaptation on both financial performance (Model 5a: $\beta = 0.20, p < 0.05$), and non-financial performance (Model 5b: $\beta = 0.22, p < 0.01$) were significant. These results indicate a complete mediation effect, as the independent variable (IDMCs) loses all significance in its relationship to the dependent variable (performance) when the mediating variable (adaptation) is introduced. Therefore, [H4a](#) is supported.

Considering the mediating effect of product development capability for the IDMC-financial performance relationship, after the inclusion of product development capability in Model 6a in [Table 5](#), the direct effect of IDMCs became insignificant (Model 6a: $\beta = 0.088, p > 0.1$), while product development capability turned out to be a significant predictor of financial performance (Model 6a: $\beta = 0.14, p < 0.1$), thus supporting the complete mediation effect ([H5b](#)). However, in the case of non-financial performance, after the inclusion of product development capability, which turned out to be a significant predictor of non-financial performance (Model 6b: $\beta = 0.17, p < 0.05$), the direct effect of IDMCs on non-financial

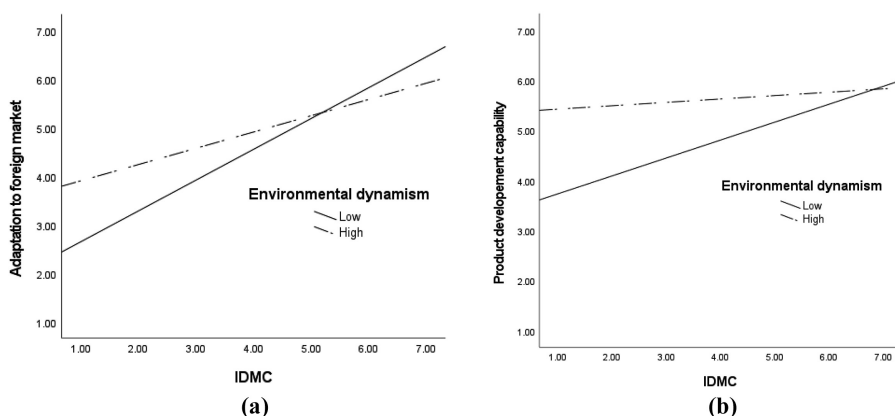


Figure 2. Moderating effects of environmental dynamism for the relationship between IDMC and (a) adaptation to foreign markets, (b) product development capability

Source(s): Authors own creation

performance remained significant ($p < 0.01$), though reduced (Model 6b: $\beta = 0.18 < \text{Model 2b: } \beta = 0.23$), thus we observe a partial mediation effect, providing support for H4b. Table 6 summarizes the results of the hypotheses testing.

6. Discussion and implications

Prior studies have explored the environmental contingencies of the relationship between generic DCs and performance, but research remains limited with respect to this relationship in extreme environmental dynamism, especially in the context of international expansion and domain-specific DCs (Colombo *et al.*, 2021). Additionally, international business and international marketing literature is overly focused on the capabilities of firms from developed economies, while the context of post-transitional countries remains somewhat overseen (Kowalik *et al.*, 2020). This study investigates the relationship between IDMCs and international performance, mediated by adaptation to foreign markets and product development capabilities and moderated by environmental dynamism.

6.1 Theoretical contributions

This study advances our understanding of the mechanisms through which domain-specific (marketing-related) dynamic capability improves international performance under conditions of extreme environmental dynamism. Accordingly, this study makes the following theoretical contributions. First, we add to prior conceptualizations of DMCs (Mitrega, 2019) by adjusting the focal construct to the international context and introducing the concept of IDMCs in a novel context. In doing so, we add to the ongoing debate on the nature of DMCs in general, and in international markets in particular. Scholarly research remains ambiguous in this regard (Mitrega, 2019), conceptualizing DMCs by focussing on: (1) resource outcomes (DMCs resulting in reconfigured marketing resources and capabilities) (Fang and Zou, 2009; Konwar *et al.*, 2017), (2) resource transformation mechanisms (DMCs as cross-functional processes) (Barrales-Molina *et al.*, 2014; Falasca *et al.*, 2017; Sheng, 2017), and (3) resource outcomes and transformation mechanisms (integrative view), whereas cross-functional processes are assumed to enable effective resource reconfiguration for marketing purposes (Buccieri *et al.*, 2020; Weerawardena *et al.*, 2015). We introduce and operationalize an extended construct of DMCs taking into account their use in international markets, referred to as IDMCs, as well as elaborating a related research model, and developing a survey instrument and testing it empirically. We conceptualize this as firms' cross-functional processes for reconfiguring marketing resources so as to create and provide customer value in international markets in response to environmental dynamism. In this regard, we empirically demonstrate that IDMCs represent higher-order constructs that operate through

Table 6.
Summary results of
hypotheses testing

Hypothesis	Relationship	Finding
H1	IDMC → Adaptation to foreign markets	Supported
H2	IDMC → Product development capability	Supported
H3	IDMC → International performance	Supported
H4a	IDMC → Adaptation to foreign markets → International performance	Supported
H4b	IDMC → Product development capability → International performance	Supported*
H5a	IDMC × Environmental Dynamism → International performance	n.s
H5b	IDMC × Environmental Dynamism → Adaptation to foreign markets	Supported
H5c	IDMC × Environmental Dynamism → Product development capability	Supported
Note(s): *For non-financial performance as DV, partial mediation effect		
Source(s): Authors' own creation		

the lower-order constructs of adaptation to foreign markets and product development capabilities, which in turn contribute to international performance. Specifically, our two-step theoretical approach moves beyond the simple linkage between domain-specific DCs and performance, and opens up the black box of IDMCs. Specifically, our conceptual model assumes that adaptation to foreign markets and product development capability are mediating variables between IDMCs and firm performance. This therefore provides scholars with additional insights into the black box of IDMCs, and determines how marketing capabilities can be leveraged so that they contribute to firm international performance (Mitreġa *et al.*, 2021). In this sense, our findings resonate and extend those of Danneels (2011) and provide additional insights into various modes of alteration to ordinary resources and capabilities. In particular, we propose that adaptation to foreign markets and product development capability are specific examples of leveraging existing resources (products) and capabilities (adaptation and product development) that can be applied to alternative uses, such as related or new product categories in foreign markets (product-market diversification) (Danneels, 2011). In particular, product adaptation represents tacit firm-specific knowledge that allows for a better understanding of international customer needs, and thus anticipation of and response to host market changes (Hollender *et al.*, 2017). In this sense, second-order international marketing competences (IDMCs) allow international customer and product knowledge to be accessed and applied to product adaptation and product development. However, more research is needed in order to understand specifically how effective this alteration mode is for foreign expansion, since resources vary in their degree of fungibility (Danneels, 2011). We therefore provide empirically grounded insights in a unique research setting into the cross-functional nature of IDMCs. Taken together, our results align conceptually with the integrative view of DMCs (Buccieri *et al.*, 2020; Pavlou and El Sawy, 2011) and the widely accepted view of the nature and consequences of domain-specific DCs (Helfat and Martin, 2015; Schilke *et al.*, 2018).

Second, this study contributes to the debate on the DC–performance relationship contingent upon environmental dynamism level, by offering a nuanced view of this relationship and considering specific-domain DCs in the international context (IDMCs), as opposed to generic DCs (Hine *et al.*, 2014). Prior studies suggest that generic DCs are effective in improving performance under conditions of environmental dynamism (Fainshmidt *et al.*, 2016). Moreover, existing research findings also suggest a positive effect of DMCs on international performance (Buccieri *et al.*, 2020). However, little is known about this relationship under conditions of extreme environmental dynamism (manifested by the economic consequences of the COVID-19 pandemic). Our study offers a new perspective by testing this relationship empirically in the unique context of a post-transformational economy and during an environmental jolt, on a sample of 277 Polish export manufacturers. In doing so, we adopt a configurational approach to the DC-performance relationship (Fainshmidt *et al.*, 2019; Wilden *et al.*, 2016), that is by including both organizational and external factors in our analysis. We use hierarchical multiple regression analysis to test our hypotheses and find support for the existing conceptualizations, except for one. The data demonstrates a significant and positive association between IDMCs and international performance (direct effect) and between IDMCs and two mediating variables – adaptation to foreign markets and product development capability, as well as between the mediators and international performance (indirect effect of IDMCs). When environmental dynamism is included as a moderator to the relationship between IDCMs and international performance, the direct effect remains significant, while the moderating effect of environmental dynamism is not supported. The significant and positive indirect effect of IDMCs on international performance (through mediators, i.e. adaptation and product development capability) is, however, weakened when companies experience a higher level of environmental dynamism. Taken together, these show the boundary conditions for the effectiveness of specific DCs (IDMCs) in

this context, a phenomenon suggested in some prior empirical studies (e.g. [Schilke, 2014](#)). In this sense, our findings extend [Mitreġa et al.'s \(2021\)](#) study of Polish exporters, which found a positive effect of DMCs on international performance (product and market expansion). From a broader perspective, we may speculate that our results contrast with prior accumulated findings about the high value of DCs in highly dynamic environments ([Fainshmidt et al., 2016](#)), indicating the boundary of the effectiveness of DCs ([Verona and Zollo, 2011](#)), which ends under conditions of extremely high dynamism. A possible explanation for this relates to a limitation of our study, i.e. that we captured companies specific actions over a short period of time. As indicated in the literature, such resource commitments may not be visible in the short term, due to insufficient opportunities to cover the cost of their development ([Drnevich and Kriauciunas, 2011](#); [Helfat et al., 2007](#)), or simply because the enterprises in our sample may have developed an irrelevant set of resources ([Wang and Ahmed, 2007](#)).

Third, our study extends prior research by focussing on a unique sample of export manufacturers from the post-transition CEE economy of Poland. In this respect, [Fainshmidt et al. \(2016\)](#) suggest that emerging and transformational economies constitute a natural context conducive to the high effectiveness of DCs. However, our study shows that under conditions of an extremely high level of environmental dynamism, seeking to employ IDMCs to create and deliver value to foreign customers has its limitations for resource-constrained companies from Poland (see [Caputo et al., 2016](#); [Dikova et al., 2016](#); [Hoskisson et al., 2013](#); [Jaklič et al., 2018](#)). A possible explanation for the weakening effect of environmental dynamism is that it is amplified by the research context itself, i.e. the economy of Poland. This finding suggests that IDMCs alone are not sufficient in order to effectively compete in international markets.

6.2 Practical implications

The study also has several practical implications for managers and policymakers in helping them to better understand how IDMCs affect international performance during discontinuous environmental changes, which is crucial for firms' survival. First, the study's findings provide suggestions about the international strategy and tactics required to improve performance in turbulent times. Namely, as the international environment changes dynamically, resulting in the decline of some industries, the importance of the use of IDMCs becomes evident. IDMCs allow for product extensions, either into related product categories, drawing on existing resources and competencies to serve particular customer segments, or through improved product adaptations. Both require constantly upgraded international knowledge and skills related to branding, promotion, distribution and product attributes, supported by sensing, seizing and reconfiguring IDMCs. By focussing on the activities mentioned, managers are better able to build resilience in international markets during turbulent times.

Second, the results may also be helpful for policymakers in designing effective support mechanisms for exporters originating from post-transitional economies during periods of environmental jolts or atypical crises. Our study demonstrates that public support should consider the time horizon and resource availability. Moreover, public policy should also consider the systemic competitive disadvantage of firms from post-transitional economies relative to their counterparts from developed economies. This is due to resource scarcity and lower technological advancement. It should be noted that the economy of Poland is dominated by firms operating in industries of relatively low technological intensity that demonstrate a generally low level of innovation ([WIPO, 2021](#)), implying their weak competitiveness in international markets. The most immediate support should therefore be directed towards providing financial resources, followed by more demanding innovation-oriented support programmes in the longer term leading to increased innovation outputs, and

hence high value-added growth. In this context, the nature of the industry should also be considered. Namely, export manufacturers are characterized by a lower potential for rapid and significant resource reconfiguration in the short term, as well as greater resource needs than service-based companies. As a whole, this study offers value to governmental bodies seeking to effectively support exporters, by indicating that public support should consider specific industry characteristics, as well as maintain a careful balance of trade-offs between short- and long-term investments and outcomes during environmental jolts.

6.3 Limitations and future research

Although this study broadens our understanding of the nature and effectiveness of IDMCs under conditions of extreme environmental turbulence, it suffers from several limitations. First, the majority (62%) of the firms in our sample were exporting under their own brand, whereas in the entire population of exporting fully Polish-owned firms, the majority are still in the process of improving their position in GVCs, hence they offer intermediate products. Our findings may also be biased due to a domination of firms operating in either the business-to-business sector or the business-to-business and business-to-customer sectors simultaneously. Another limitation pertains to the cross-sectional nature of our study, which provides a snapshot of a longer process, whose ultimate results may only be visible after another few years. Consequently, we encourage researchers to study the effectiveness of IDMCs in other institutional and country contexts during the pandemic using longitudinal qualitative and quantitative research approaches, such as case studies and panel datasets, so as to fully understand the significance of extreme environmental dynamism as a boundary condition for the effectiveness of DCs.

Our findings shed light on potential avenues for future research. First, we call for further research that allows for a better understanding of the relationship between IDMCs and performance under conditions of low, moderate and high environmental dynamism, as well as in other economic contexts. In this vein, it would be interesting to explore whether and how the presence (or lack) of developed external institutions are expected to moderate the relationship between IDMCs, other internal factors and international performance. Second, another potentially fruitful avenue for research into the effectiveness of IDMCs may be exploring additional internal factors that may be significant in foreign expansion, such as managerial risk aversion, dynamic managerial capabilities (Helfat and Peteraf, 2015) and psychic distance. Third, we also encourage researchers to include export-oriented companies from service industries in order to reveal the effectiveness of generic DCs, IDMCs or other specific-domain DCs during extreme environmental dynamism. We might speculate that decisions on firms' resource reconfiguration should consider sequential targeting of marketing programme elements. For instance, while potential returns from price and promotion changes might be rapidly achieved at relatively low cost, the potential benefits from investments in product and geographical markets, manifested by R&D programmes, are likely delayed in time, but in the long term improve profitability and a firm's performance. However, such initiatives may also threaten short-term profitability and a firm's existence, as too radical resource reconfiguration might be harmful (see Ambrosini *et al.*, 2009; Wenzel *et al.*, 2020). In this respect, development of IDMCs may ensure a balance between short-term gains and long-term growth in foreign markets. We therefore invite other scholars to study in more detail the role of sequencing of marketing-mix elements as a result of development and deployment of IDMCs, and its impact on performance.

Overall, we call for further research to extend our findings using other methodologies, larger samples and other country contexts. We are aware that the spread of the pandemic from early 2020 constituted rare natural experimental conditions, but from the current perspective, the occurrence of other environmental jolts in the foreseeable future is both more

possible and most likely more frequent than in past decades. Understanding of the effectiveness of IDMCs and the mediating mechanisms through which they affect international performance is of great importance to exporters, as applying them contributes to an improvement in international competitiveness.

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